

The reference habitat network

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1 Overview

SPEC section 45 fixes a small ecological Bayesian network as the integration test of the whole ecosystem:

```
Climate -> SoilMoisture -> Vegetation -> HabitatQuality -> Occupancy
Irrigation -> SoilMoisture
GrazingPressure -> Vegetation
```

This vignette runs it end to end: the model from the zoo, its picture, validation, the brute-force joint distribution against exact inference, evidence, the difference between observing and intervening on grazing pressure, and the same answers from the Netica and GeNIe files that ship with `BayesianNetworkFormats.jl`.

The network is deliberately in the shape the ecological guidelines ask for ([Marcot et al. 2006](#); [Chen and Pollino 2012](#)): a small number of discretised variables, each arc a stated mechanism, and a management lever separated from the quantity it acts on – which is also where the discretisation and elicitation costs discussed by Uusitalo ([2007](#)) and Kuhnert et al. ([2010](#)) would fall in a real study.

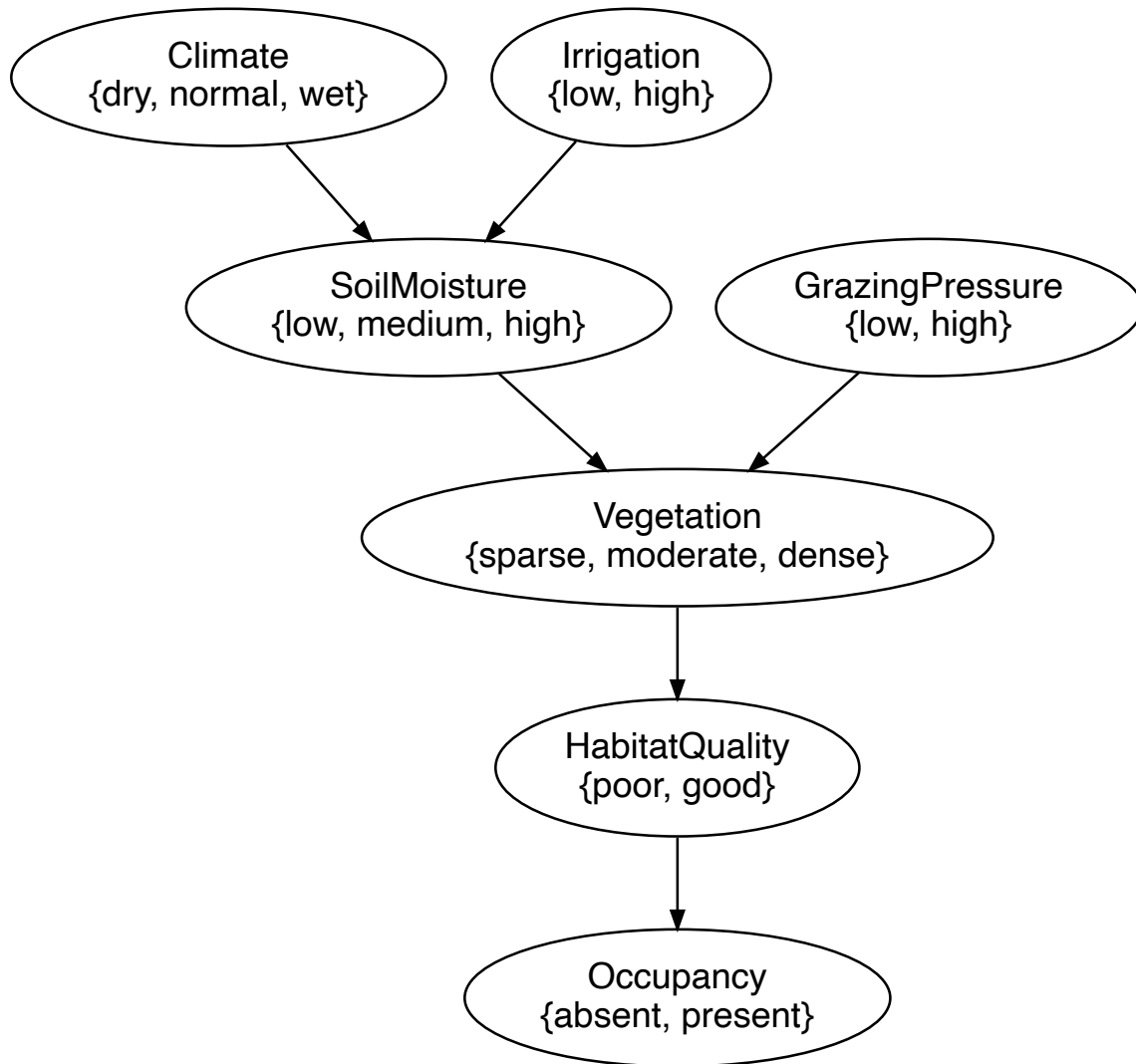
2 Setup

```
using EcologicalBayesianNetworks
using BayesianNetworks
using BayesianNetworkInference
m = load_model("reference_habitat_bn")
```

BayesModel(7 variables, 7 mechanisms, 7 kernels)

`load_model` returns `BayesianNetworks.reference_habitat_model()`, the structure of `reference_habitat_bn()` with the conditional probability tables of the `habitat_reference` fixtures bound. Chance variables are ellipses; every arc is a mechanism input.

```
to_graphviz(m)
```



```
model_summary("reference_habitat_bn")
```

```

ModelSummary reference_habitat_bn (Reference habitat network (SPEC section 45))
format / licence      julia / MIT (builtin)
nodes                 7 (7 chance, 0 decision, 0 utility)
arcs                  6
largest state space   3
largest in-degree     2
load_model            BayesModel
  
```

3 Validation

`validate` checks the syntax (one mechanism per variable, acyclic, contiguous positions) and, with `semantics = true`, that every mechanism resolves to a kernel whose domain and codomain

match its parents and target and that every row sums to one. It returns nothing when everything is in order.

```
validate(m; closed = true, unique_names = true, semantics = true)
```

4 Joint distribution versus exact inference

With seven variables the joint distribution has only 432 entries, so the reference semantics of SPEC section 14, the product of all the kernels, can be enumerated outright. `joint_distribution` returns it as a state $I \rightarrow (x)$ variables.

```
J = joint_distribution(m)
size(J.table), sum(J.table)
```

```
((3, 2, 3, 2, 3, 2, 2), 1.0000000000000002)
```

`marginal` sums the joint; `BayesianNetworkInference.infer` answers the same query by variable elimination without ever forming the joint. The two agree to rounding.

```
p_occ = marginal(m, :Occupancy)
```

```
FiniteKernel{Float64}(I → Occupancy{absent,present})
  ()
  absent  0.5238
  present 0.4762
```

```
post, diag = infer(m, :Occupancy)
post.table ≈ p_occ.table, diag
```

```
(true, InferenceDiagnostics([:Climate, :Irrigation, :SoilMoisture, :GrazingPressure,
:Vegetation, :HabitatQuality], 18, 6, 2))
```

The junction-tree backend gives every marginal from one calibration:

```
ms = all_marginals(m; backend = JunctionTree())
round.(ms[:Vegetation].table; digits = 4)
```

```
3-element Vector{Float64}:
 0.3611
 0.3798
 0.259
```

Ancestral sampling (SPEC section 18) draws from the same distribution; the empirical frequency of occupancy converges to the exact value.

```
s = ancestral_sample(m, 20_000)
round.(empirical_marginal(s, :Occupancy).table; digits = 3)
```

```
2-element Vector{Float64}:
 0.524
 0.476
```

5 Evidence

Evidence conditions the joint. `posterior` reads the numbers off as a dictionary; `observe` records the evidence on the model so that every later query uses it.

```
posterior(m, :Occupancy; evidence = Dict{:Vegetation => :dense}))
```

```
Dict{Symbol, Float64} with 2 entries:
 :present => 0.6675
 :absent  => 0.3325
```

```
me = observe(m, :Vegetation => :dense)
marginal(me, :Occupancy).table ≈ infer(me, :Occupancy)[1].table
```

```
true
```

6 Observing versus intervening

SPEC section 45 asks for $P(\text{Occupancy} = \text{present} \mid \text{do}(\text{GrazingPressure} = \text{low}))$. A hard intervention rewrites the mechanism of `GrazingPressure` into the point mass at `low` (SPEC sections 21 and 22); observation conditions on the event instead. The two operations are recorded differently: the intervention appears in the syntax and the history, the observation in the evidence.

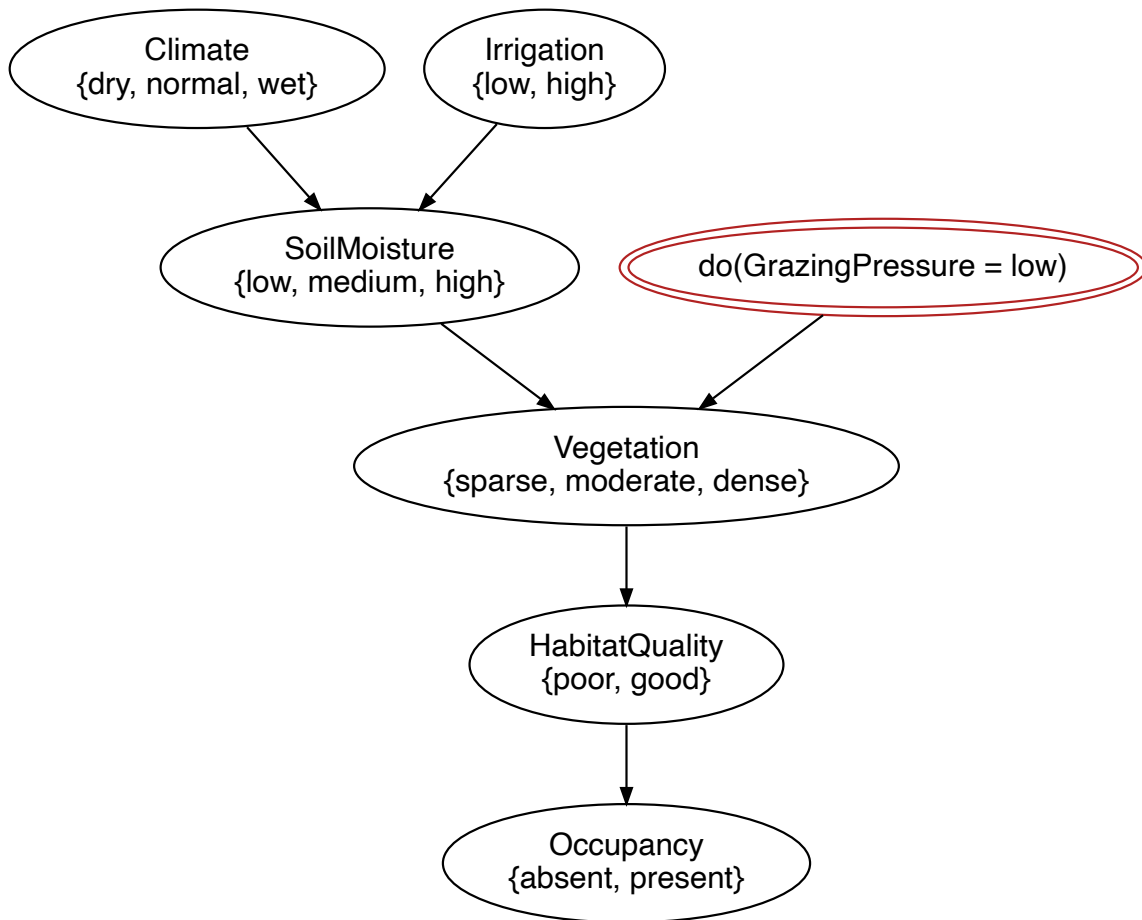
```
low_do = do_intervention(m, :GrazingPressure => :low)
low_obs = observe(m, :GrazingPressure => :low)
(intervene = marginal(low_do, :Occupancy).table[2], observe = marginal(low_obs,
:Occupancy).table[2])
```

```
(intervene = 0.51166025, observe = 0.51166025)
```

```
# The raw events carry a wall-clock timestamp, which would make this committed
# output differ on every render; show what the intervention did instead.
[(e.kind, e.target, e.added.kernel_ref) for e in history(low_do)]
```

```
1-element Vector{Tuple{Symbol, Symbol, PointMassRef}}:
 (:hard, :GrazingPressure, PointMassRef(:low))
```

```
to_graphviz(low_do)
```



Because GrazingPressure is a root, the two numbers coincide: nothing upstream can be learned from it. On a variable with parents the difference shows. Observing dense vegetation is evidence that the soil was moist; forcing dense vegetation tells us nothing about the soil, because the arc from SoilMoisture has been cut.

```
(intervene = round.(marginal(do_intervention(m, :Vegetation => :dense),  
:SoilMoisture).table; digits = 4),  
observe = round.(marginal(observe(m, :Vegetation => :dense), :SoilMoisture).table;  
digits = 4),  
prior = round.(marginal(m, :SoilMoisture).table; digits = 4))
```

```
(intervene = [0.288, 0.403, 0.309], observe = [0.0834, 0.35, 0.5666], prior =  
[0.288, 0.403, 0.309])
```

Downstream of the rewritten variable the two agree, since Occupancy depends on the rest of the network only through Vegetation:

```
marginal(do_intervention(m, :Vegetation ⇒ :dense), :Occupancy).table ≈
    marginal(observe(m, :Vegetation ⇒ :dense), :Occupancy).table
```

```
true
```

7 The same model from Netica and GeNIe files

The `habitat_reference` fixtures of `BayesianNetworkFormats.jl` hold the same network in Netica `.dne` and GeNIe `.xdsl` form. `read_bayesnet` parses either into a `BayesModel`, and the intervention query gives identical answers from all three sources.

```
dne = read_bayesnet(fixture_path("dne/habitat_reference.dne"))
xdsl = read_bayesnet(fixture_path("xdsl/habitat_reference.xdsl"))
dne ≈ m, xdsl ≈ m
```

```
(true, true)
```

```
query(model) = marginal(do_intervention(model, :GrazingPressure ⇒ :low),
    :Occupancy).table[2]
query(m), query(dne), query(xdsl)
```

```
(0.51166025, 0.51166025, 0.51166025)
```

The zoo's `model_ir` gives the `NetworkIR` of the Julia-built model, which `write_network` can turn back into any of the file formats.

```
ir = model_ir("reference_habitat_bn")
[(v.id, v.parents) for v in ir.variables]
```

```
7-element Vector{Tuple{Symbol, Vector{Symbol}}}:
 (:Climate, [])
 (:Irrigation, [])
 (:SoilMoisture, [:Climate, :Irrigation])
 (:GrazingPressure, [])
 (:Vegetation, [:SoilMoisture, :GrazingPressure])
 (:HabitatQuality, [:Vegetation])
 (:Occupancy, [:HabitatQuality])
```

8 Summary

The SPEC section 45 reference network is small enough that exact inference can be checked against the brute-force joint distribution entry by entry, which is what makes it the integration test of the whole ecosystem. Observing vegetation can update the upstream soil moisture, whereas intervening on vegetation leaves soil moisture at its prior. `GrazingPressure` is a root in

this example, so observing it updates nothing upstream. Downstream agreement requires the relevant screening-off conditions; it is not a general consequence of being downstream. The API keeps observation and intervention distinct. The same model read from the Netica and GeNIe files gives the same numbers. The next vignette, *The management influence diagram*, adds a decision and two utilities to this network.

9 References

- Chen, Serena H., and Carmel A. Pollino. 2012. “Good Practice in Bayesian Network Modelling.” *Environmental Modelling & Software* 37: 134–45. <https://doi.org/10.1016/j.envsoft.2012.03.016>.
- Kuhnert, Petra M., Tara G. Martin, and Shane P. Griffiths. 2010. “A Guide to Eliciting and Using Expert Knowledge in Bayesian Ecological Models.” *Ecology Letters* 13 (7): 900–914. <https://doi.org/10.1111/j.1461-0248.2010.01477.x>.
- Marcot, Bruce G., J. Douglas Steventon, Glenn D. Sutherland, and Robert K. McCann. 2006. “Guidelines for Developing and Updating Bayesian Belief Networks Applied to Ecological Modeling and Conservation.” *Canadian Journal of Forest Research* 36 (12): 3063–74. <https://doi.org/10.1139/x06-135>.
- Uusitalo, Laura. 2007. “Advantages and Challenges of Bayesian Networks in Environmental Modelling.” *Ecological Modelling* 203 (3–4): 312–18. <https://doi.org/10.1016/j.ecolmodel.2006.11.018>.